Strategic Workforce Optimization Using HR Analytics

Business Analysis
SQL
August 2026

ABOUT The Client

The Client, is a fast-growing technology company, that is experiencing increasing concerns around employee turnover, inconsistent performance tracking, and unclear salary structures across departments. While the company has succeeded in attracting top talent, leadership lacks a consolidated, data-driven view of workforce health , which is making it difficult to identify why employees are leaving, which departments are underperforming, and whether compensation is fairly aligned with performance.

Key Analytical Findings

9 & 9

Highest turnover

Engineering & Marketing

4.12 / 4.07

Performance: left / stayed

Leavers scored higher

4.00

Lowest dept. performance

Sales

$80K–$83K

Salary range

Flat across departments

22 days

Top absence case

John Green (of 5 flagged)

Query | Query History

```sql
select * from attendance;
select * from department;
select * from employee;
select * from performance;
select * from salary;
select * from turnover;


/* 1. Which department has the most employees leaving? */

select d.department_name, count(t.turnover_id) as No_of_Leavers
from department as d
join turnover as t on d.department_id = t.department_id
group by d.department_name
order by No_of_Leavers desc;

```

Scratch Pad

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

	department_name character varying (30)	no_of_leavers bigint
1	Engineering	9
2	Marketing	9
3	HR	6
4	Sales	4

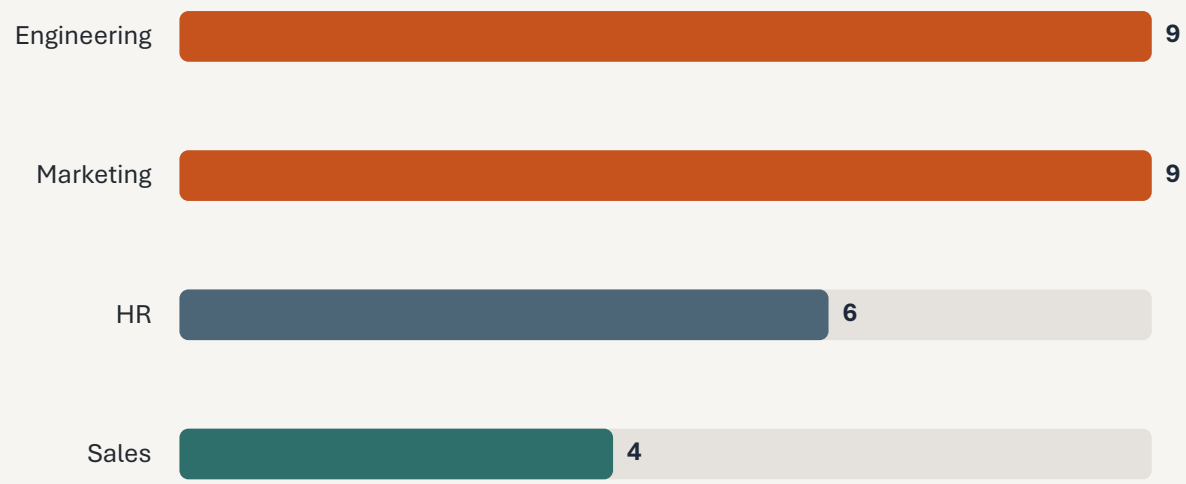
Insight 01 — Turnover is concentrated in Engineering and Marketing

Question: Which department has the most employees leaving?

9
Engineering & Marketing exits

6
HR exits

4
Sales exits



What this means

- Engineering and Marketing are tied for the highest number of leavers
- The four departments total 28 recorded leavers in this result set; 18 are in Engineering or Marketing.
- Retention attention should start with these two departments, while Sales shows the fewest departures.

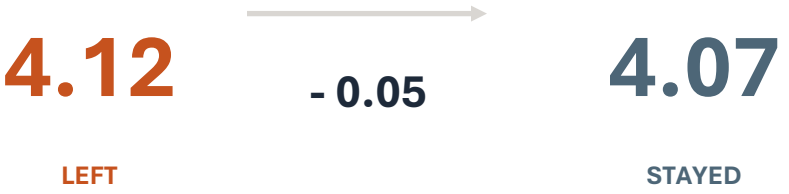
```
33
34
35  /* 2. What is the average performance score for employees who left vs. employees who stayed? */
36
37  select (case when t.turnover_date is not null then 'left'
38              else 'stayed' end) as employee_status, Round (avg(p.performance_score) :: numeric, 2) as avg_performance_score
39  from employee e
40  left join turnover t on e.employee_id = t.employee_id
41  join performance p on e.employee_id = p.employee_id
42  group by employee_status;
43
44  /* select count(e.employee_id),sum(p.performance_score),(case when t.turnover_date is not null then 'left'
45              else 'stayed' end) as employee_status, avg(p.performance_score) as avg_performance_score
46  from employee e
47  left join turnover t on e.employee_id = t.employee_id
48  join performance p on e.employee_id = p.employee_id
49  group by employee_status; */
50
```

	employee_status text	avg_performance_score numeric
1	left	4.12
2	stayed	4.07

Insight 02 — Leavers are not the lowest-performing group

Question: What is the average performance score for employees who left vs. employees who stayed?

Average performance score



What this means

- The average score for leavers is 4.12 versus 4.07 for stayers.
- Leavers averaged 4.12 vs. 4.07 for those who stayed, a small but meaningful signal that The Client may be losing capable employees, not just underperformers.
- Retention analysis should therefore examine factors beyond performance score alone.

```
50
51
52
53  /* 3. Which department has the lowest average performance score? */
54
55  select d.department_name, Round  (avg(p.performance_score) :: numeric, 2)
56  from department as d
57  join performance as p on d.department_id = p.department_id
58  group by d.department_name
59  order by avg(p.performance_score);
60
61  /* select d.department_id, d.department_name, count(p.department_id),sum(p.performance_score),avg(p.performance_score)
62  from department as d
63  join performance as p on d.department_id = p.department_id
64  group by d.department_id, d.department_name
65  order by avg(p.performance_score) desc; */
66
```

	department_name character varying (30)	round numeric
1	Sales	4.00
2	HR	4.05
3	Engineering	4.10
4	Marketing	4.13

Total rows: 4 | Query complete 00:00:00.094 | CRLF | Ln 60, Col 1

Insight 03 — Sales has the lowest average performance score

Question: Which department has the lowest average performance score?



4.00

Lowest department average

Sales

The spread is narrow: 0.13 points from lowest to highest.

What this means

- Sales ranks lowest on average performance at 4.00.
- Marketing is highest at 4.13; the overall department range is tight.
- The result suggests a targeted performance review for Sales and also a broad workforce capability check.



Query | Query History

Scratch Pad

```
67
68  /* 4. What is the average salary paid in each department? */
69
70  select d.department_name, Round (avg(s.salary_amount):: numeric, 2) as avg_salary
71  from department as d
72  join salary as s on d.department_id = s.depaartment_id
73  group by d.department_name;
74
75
76
77  /* select d.department_id, d.department_name, count(s.salary_id),sum(s.salary_amount),avg(s.salary_amount)
78  from department as d
79  join salary as s on d.department_id = s.depaartment_id
80  group by d.department_id, d.department_name; */
81
82
83
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

	department_name character varying (30)	avg_salary numeric
1	Sales	82857.14
2	Engineering	80000.00
3	HR	83000.00
4	Marketing	80000.00

Total rows: 4 | Query complete 00:00:00.104 | CRLF | Ln 83, Col 1

Insight 04 — Average salary is relatively clustered across departments

Question: What is the average salary paid in each department?



$83k

Highest average salary

HR

$80k

Lowest average salary

Engineering + Marketing

What this means

- HR has the highest average salary at $83,000.
- Engineering and Marketing are tied at $80,000.
- The narrow range means pay differences alone does not account for the turnover pattern shown in Q1.

```
81
82
83
84  /* 5. Which employees have the highest number of absences? */
85
86  select concat(e.first_name, ' ', e.last_name) as employee_name,
87         sum(case when a.attendance_status = 'Absent' then 1 else 0 end) as No_of_absences
88  from employee as e
89  join attendance as a on e.employee_id = a.employee_id
90  group by e.employee_id
91  order by No_of_absences desc
92  limit 5;
93
94
95
96
97
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1 of 1

	employee_name text	no_of_absences bigint
1	John Green	22
2	Charlie Smith	21
3	John Wilson	21
4	John Brown	21
5	Frank Wilson	20

Total rows: 5 | Query complete 00:00:00.105 | CRLF | Ln 94, Col 1

Insight 05 — A small group has the highest absence counts

Question: Which employees have the highest number of absences?

Employee	Absences
John Green	22
Charlie Smith	21
John Wilson	21
John Brown	21
Frank Wilson	20

22 absences highest individual count in the displayed output

What this means

- John Green has the highest absence count at 22.
- The top five range from 20 to 22 absences, indicating a closely grouped high-absence cluster.
- Attendance flags are best treated as a follow-up signal for manager review, not a standalone explanation for turnover.

Overall analysis : the strongest signal is turnover concentration

Across the five SQL questions, the most actionable pattern is departmental concentration rather than one single employee attribute.

Turnover

Engineering + Marketing

18 of 28 recorded leavers

Performance

Sales lowest

4.00 average

Salary

HR highest

~$83k average

Attendance

Top 5 clustered

20–22 absences

Interpretation The output suggests targeted departmental action, with performance and attendance used as supporting signals rather than assumed root causes.

Management priorities — focus resources where the signals overlap

1	**Stabilize Engineering + Marketing**	Highest recorded leaver counts; start retention diagnostics here.
2	**Review Sales performance conditions**	Lowest department average performance at 4.00; assess coaching, workload, and role fit.
3	**Use compensation as a context variable**	Salary levels vary modestly; compare pay patterns alongside turnover rather than in isolation.
4	**Review high-absence cases**	Use the top-five attendance list as a manager follow-up queue, with context and consistency checks.

Final recommendations

Turn the findings into a focused workforce action plan.

01	**Launch retention reviews in Engineering and Marketing**	Identify why exits cluster there through manager interviews, exit reasons, workload review, and role-level cuts.
02	**Create a targeted Sales performance support plan**	Use coaching and capability diagnostics around the 4.00 department average; track progress over time.
03	**Treat pay as a reason not as sole-driver**	Review salary positioning with turnover and performance together before changing compensation policy.
04	**Set an attendance follow-up workflow**	Review the five highest absence counts with appropriate context and also perform Individual wellbeing check-ins and workload review for the five flagged employees and also the rest of the employees with absences.

THANK YOU

Hope to get Feedback.